Applied Economics

OLS Variance and Robust Standard Errors

Pello Aspuru

Winter 2025/26

Plan for this Lecture

How well does random variable $\hat{\beta}$ represent the number β?

- Last session: $\hat{\beta}$'s Unbiasedness, consistency, asymptotic distribution.
- This session: The variance of $\hat{\beta}$.

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Variance of the OLS Estimator

Variance of OLS Estimator $\hat{\beta}$

Economic question: does pollution X_{1i} cause a decrease of housing prices Y_i in Vienna?

Model for observed sample n: $Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{u}_i \quad i = 1, ...n \quad k = 2$

```
n <- 100
df <- population[sample(nrow(population), n), ]
ols_est = lm(price ~ nox + rooms, data = df)
```

The uncertainty of $\underset{(k+1\times 1)}{\hat{\beta}}$ is represented by variance-covariance matrix:

$$\underset{(k+1\times k+1)}{Var(\hat{\beta}|\mathbf{X})} = \begin{pmatrix} Var(\hat{\beta}_0|\mathbf{X}) & cov(\hat{\beta}_0, \hat{\beta}_1|\mathbf{X}) & \dots & cov(\hat{\beta}_0, \hat{\beta}_k|\mathbf{X}) \\ cov(\hat{\beta}_0, \hat{\beta}_1|\mathbf{X}) & Var(\hat{\beta}_1|\mathbf{X}) & \dots & cov(\hat{\beta}_1, \hat{\beta}_k|\mathbf{X}) \\ \vdots & \vdots & \ddots & \vdots \\ cov(\hat{\beta}_0, \hat{\beta}_k|\mathbf{X}) & cov(\hat{\beta}_1, \hat{\beta}_k|\mathbf{X}) & \dots & Var(\hat{\beta}_k|\mathbf{X}) \end{pmatrix}$$

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Variance of OLS Estimates $Var(\hat{\beta}|\mathbf{X})$

From last week's unbiasedness derivtion, $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\underset{(n\times 1)}{\mathbf{u}}$

$$\underset{(k+1\times k+1)}{Var(\hat{\beta}|\mathbf{X})} = Var(\hat{\beta} - \beta|\mathbf{X}) = E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)'|\mathbf{X}]$$

$$= E[((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u})((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u})'|\mathbf{X}]$$

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- What do we observe? What we do not observe?

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X <- as.matrix(df[,names(df) %in% c("constant","nox", "rooms")])
Y <- as.matrix(df[,names(df) %in% c("price")])
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- Next step: unpack $\underset{n\times n}{E[\mathbf{u}\mathbf{u}'|\mathbf{X}]}$

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Variance of the error terms $Var(\mathbf{u}|\mathbf{X})$

Last session's **Assumption 3: Random sampling**: $E[u_i u_j] = 0$ for all $i \neq j$:

$$E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \begin{pmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n^2 \end{pmatrix} = diag(\sigma_1^2, ..., \sigma_n^2) \equiv \Omega$$

- Approximate each unknown σ_i^2 with $\hat{u}_i Y_i - \mathbf{X_i}\hat{\beta}$.

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beta_hat = solve(t(X) %*% X) %*% t(X) %*% Y
u_hat_sq = (as.vector(Y - X %*% beta_hat))^2
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- Next, construct $\widehat{\Omega} = diag(\hat{u}_1^2, \hat{u}_2^2, ..., \hat{u}_n^2) = diag(\hat{u}^2)$ Where $\underset{n\times 1}{\hat{u}} = \mathbf{Y} - \mathbf{X}\hat{\beta}$

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Robust Variance Estimator

Plugging $\widehat{\Omega}$ into the formula for $Var(\hat{\beta}|\mathbf{X})$ yields a consistent estimator for the variance matrix of the OLS estimator:

$$\widehat{Var}(\hat{\beta}|\mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\hat{\Omega}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$$

```
var_beta_hat <-
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- Diagonal elements of $\widehat{Var}(\hat{\beta}|\mathbf{X})$: estimated variances of parameter estimates.
- Taking the square roots yields the **standard errors** $\widehat{SE}(\hat{\beta}_j)$ $\rightarrow$ hypothesis testing.

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SE_hat <- sqrt(diag(var_beta_hat))
```

$\rightarrow$ How do $\widehat{SE}(\hat{\beta}_j)$ compare tho those reported by the lm() function?

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Heteroskedasticity and Homoskedasticity

Why the different standard error computation prodcedures gave different results?

- $\widehat{Var}(\hat{\beta}|\mathbf{X})$ formula-based standard errors: **heteroskedasticity-robust**.
- lm() function standard errors: **non-robust**, assume **homoskedasticity**.

Assumption 5: Conditional Homoskedasticity

All errors errors u_i feature the same conditional variance σ_u^2:

$$var(u_i|\mathbf{X_i}) = \sigma_u^2 \text{ for all } i = 1, ...n. \quad \rightarrow \quad \underset{n\times n}{var(\mathbf{u}|\mathbf{X})} = \sigma_u^2 \cdot I_n \quad \rightarrow \quad \hat{\Omega} = \hat{\sigma}_u^2 \cdot I_n$$

Under conditional homoskedasticity, OLS estimator variance simplifies to:

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Heteroskedasticity and Homoskedasticity

Assumption 5 is implausible in most economic applications.

Example: Among big houses, the price dispersion is larger than among small houses.

$\rightarrow$ Unexplained variation price (Y) depends on size (X_{2i}). If Assumption 5 is violated, the non-robust standard errors based on the simplified formula are inconsistent.

The robust standard errors are consistent under both conditional homoskedasticity and conditional heteroskedasticity:

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robust_se <- sqrt(diag(vcovHC(ols_est, type = "HC0")))
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Excursion: R^2 (goodness of fit)

R^2 is commonly reported in regression output in R, and measures what share of the total variation in the dependent variable is explained by variation in the regressors:

$$R^2 = \frac{SSE}{SST} = 1 - \frac{SSR}{SST}, \text{ with}$$

- **Total sum of squares** (SST): $\sum_{i=1}^{N}(y_i - \bar{y})^2$
- **Explained sum of squares** (SSE): $\sum_{i=1}^{N}(\hat{y}_i - \bar{y})^2$
- **Residual sum of squares** (SSR): $\sum_{i=1}^{N} \hat{u}_i^2$

The (adjusted) R^2 is sometimes used to compare different models (for the same dependent variable Y), but it does not play a major role when your goal is causal analysis.

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Multicollinearity and the OLS variance I

Does pollution (X_{1i}) cause a decrease of house prices? New control, X_{3i} property tax.

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \hat{u}_i \quad i = 1, ...n \quad k = 3$$

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ols_est_2 = lm(price ~  nox + rooms + proptax, data = df)
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R_j^2 is the R^2 from a regression of X_j on the other regressors. $\rightarrow$ measures multicollinearity (linear correlation) between X_j and the remaining regressors.

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Multicollinearity and the OLS variance II

$$\text{Higher } R_j^2 \quad \rightarrow \quad \text{higher } \hat{Var}(\hat{\beta}_j|X)$$

Intuition: If two regressors are strongly correlated, it is hard to disentangle their individual partial effects on Y.

$\rightarrow$ Do consumers dislike houses because of pollution X_{1i} or property taxes X_{3i}?

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Multicollinearity and the OLS variance II

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Drivers of estimation uncertainty

We will assume homoskedasticity for illustrative purposes, to analyse which factors drive the variance of the OLS estimator.

Variance of estimator $\hat{\beta}_j$ under homoskedasticity:

$$Var(\hat{\beta}_j|\mathbf{X}) = \frac{1}{n} \cdot \frac{\sigma_u^2}{\sigma_{X_j}^2} \cdot \frac{1}{1 - R_j^2}$$

- shrinks as sample size n increases.
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Recommended Literature

Heiß Chapters 1.9, 3.4, 8.1

Hanck et al. Chapter 5.4